

The background of the image features a dark purple gradient. On the right side, there are large, overlapping, curved shapes in shades of orange, pink, and blue, creating a dynamic, abstract design.

AWS re:Invent

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SUP308-R

Making architectural trade-offs using AWS Well-Architected Framework

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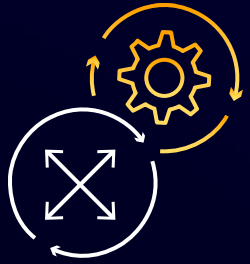
Agenda

- ❖ Introduction to the AWS Well-Architected Framework
- ❖ Well-Architected Framework optimization mechanism
- ❖ Trade-offs methodological approach
- ❖ Demo

Introduction to the AWS Well-Architected Framework



Pillars of the AWS Well-Architected Framework



Operational
excellence



Security



Reliability



Performance
efficiency



Cost
optimization



Sustainability

Well-Architected Framework optimization mechanism

Architectural optimization mechanism

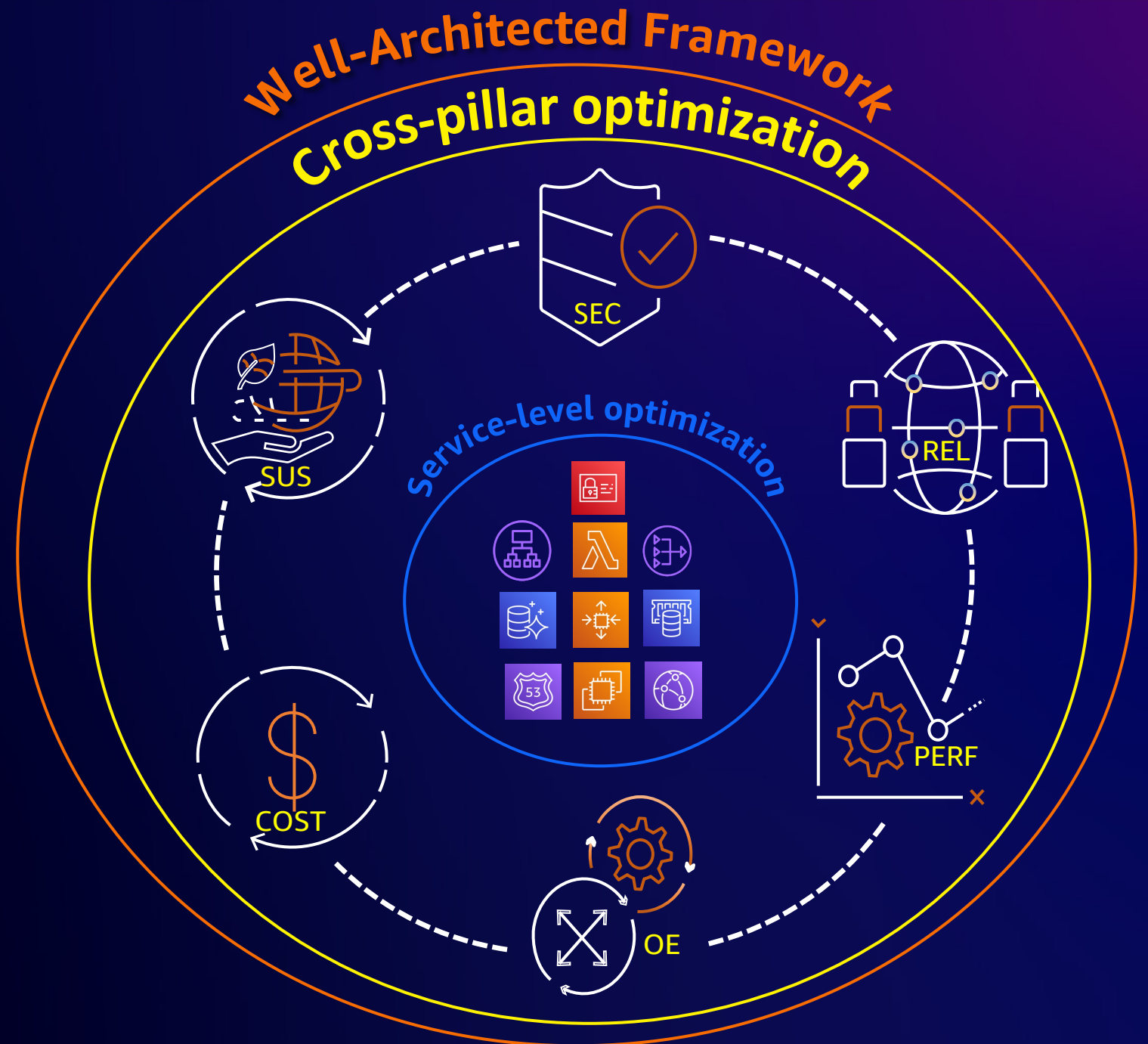


Security is top priority at Amazon



Trade-off dimensions

- Pillar level
- Best practice level
- Service level
- Code level



Trade-offs methodological approach

A semi-quantitative approach for trade-off analysis in Well-Architected Framework



$$Total\ score = \frac{W1 \times S_{i1} + W2 \times S_{i2} + W3 \times S_{i3} + W4 \times S_{i4} + W5 \times S_{i5} + W6 \times S_{i6}}{W1 + W2 + W3 + W4 + W5 + W6}$$

e. g., W_j and S_{ij} can be in range of [1,5]

Demo - Trade-off analysis in the Well-Architected Framework

Step 1

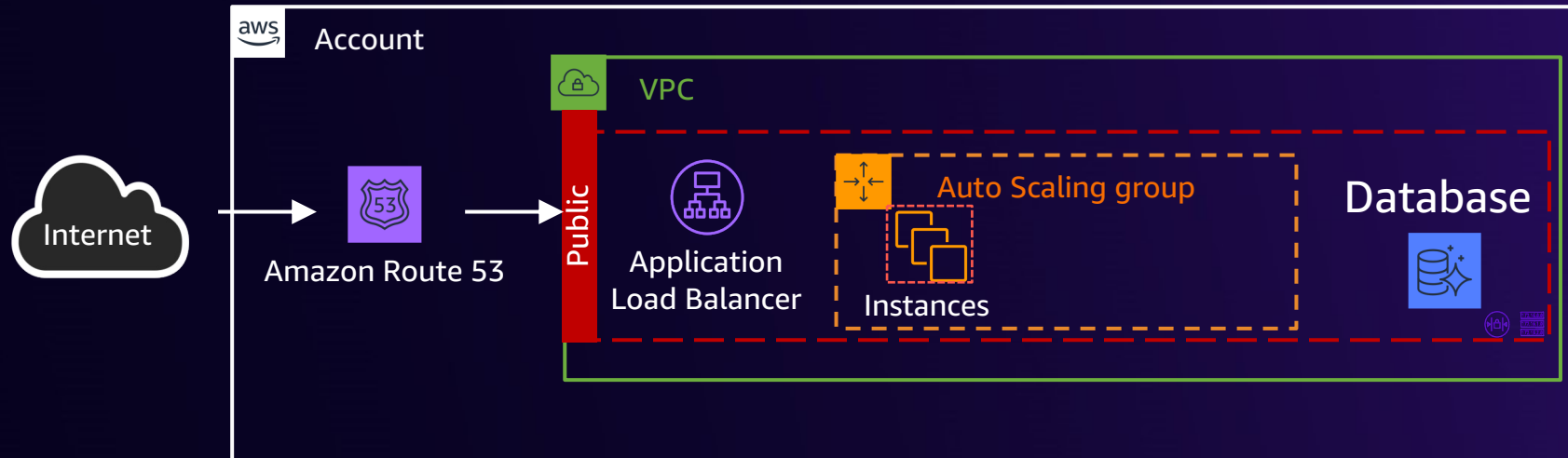
Understand and plan



Example workload: Context

Ecommerce website

- An ecommerce website is hosted on AWS
- This web application allows users to buy and sell securely on AWS
- The application needs to be available 24/7, provide a seamless user experience, and ensure data security.



Assumptions

- ❖ The workload needs security improvement.
- ❖ The workload is deployed in the same account as other company applications.
- ❖ Latency is the key performance metric for the company.

Step 2

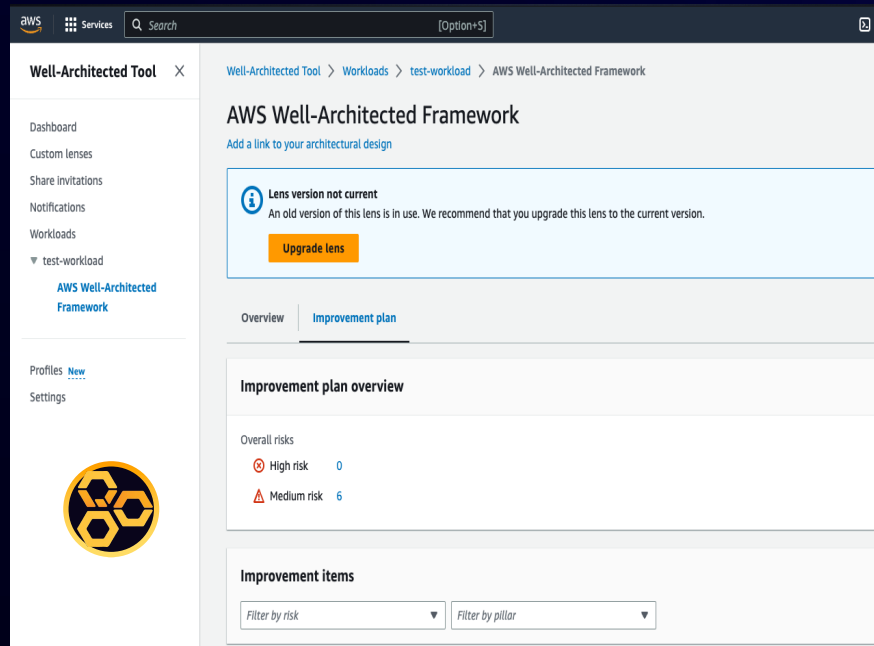
Analyze and prioritize



Improvement plans identified

- ❖ A Well-Architected Framework review has been run
- ❖ Five improvement plans have been identified

Well-Architected Tool



Improvement plans

- SEC01-BP01 Separate workloads using accounts
- SEC02-BP01 Use strong sign-in mechanisms
- SEC02-BP03 Store and use secrets securely
- SEC05-BP01 Create network layers
- SEC05-BP02 Control traffic at all layers
- SEC06-BP03 Implement managed services
- SEC06-BP04 Automate compute protection

Services to use

Option 1:
AWS Secrets
Manager

Option 2:
AWS KMS +
Custom

Option 3:
Third-party

SEC02-BP03

Store and use secrets securely



Option 1
Use AWS Secrets
Manager



Option 2
**AWS KMS and
custom solution**



Option 3
Use a third-party solution

Assumptions

- ❖ Latency is the key performance metric
- ❖ Third-party solution is cheaper compared to AWS services

Trade-off analysis

	Security (5)	Reliability (3)	Performance (3)	Cost (3)	Sustainability (1)	Ops. exe. (5)	Total score
Option 1: AWS Secrets Manager	5	5	5	4	3	5	4.75
Option 2: AWS KMS and custom	5	3	3	1	3	1	2.70
Option 3: Third party	5	3	5	5	3	5	4.60

SEC05-BP01/BP02

Create/control traffic at all network layers



Option 1
Amazon VPC



Option 3
Amazon VPC, AWS WAF, and
AWS Shield Advanced



Option 2
Amazon VPC and
AWS WAF

Assumption

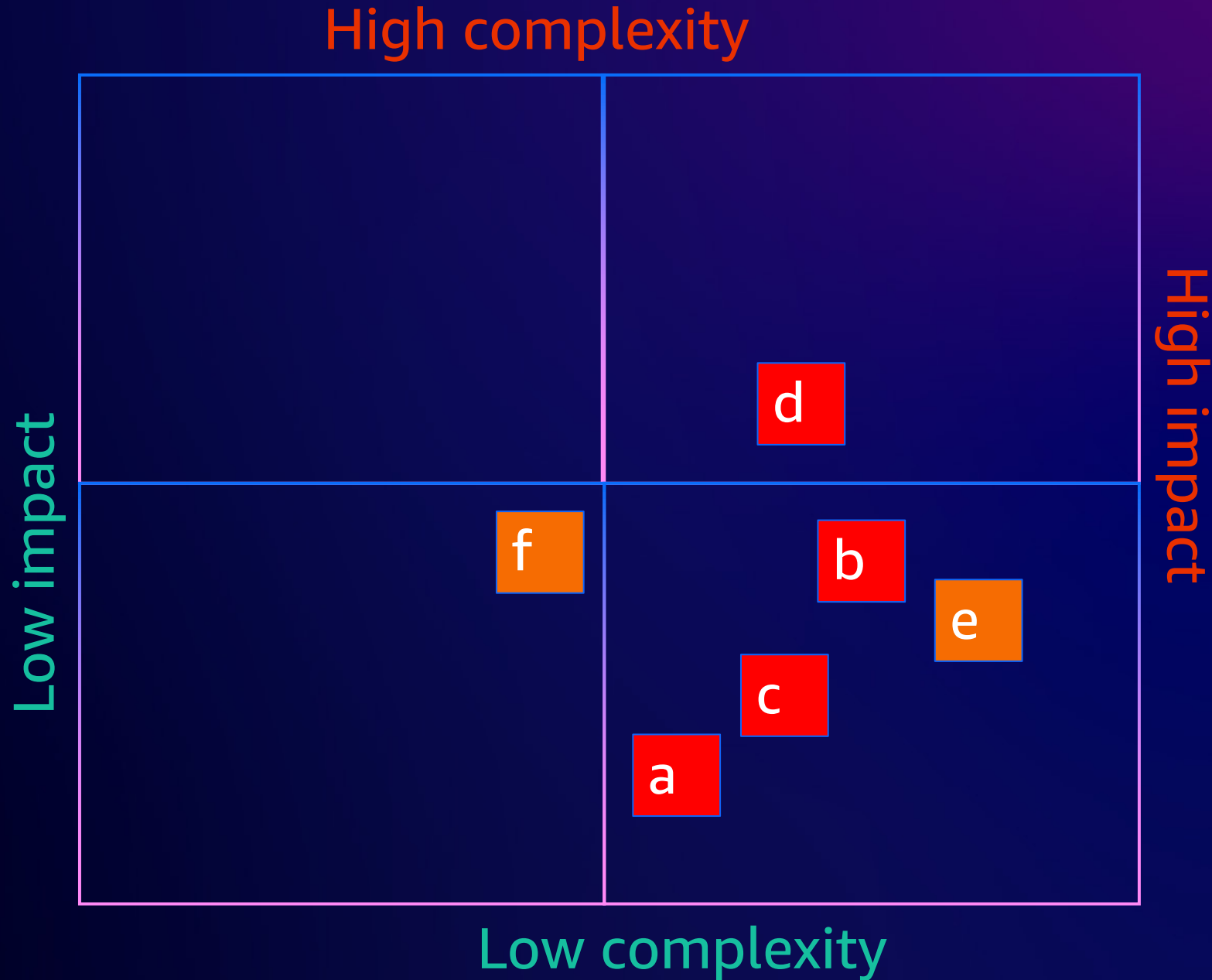
- ❖ Adding networking layers has minimal impact on latency
- ❖ VPC includes all required security components such as: public and private subnets, RT, ACL and SG, IGW

Trade-off analysis

	Security (5)	Resiliency (3)	Performance (3)	Cost (3)	Sustainability (1)	Ops. efforts (5)	Total score
Option 1: VPC	1	5	5	5	5	5	4
Option 2: VPC + WAF	4	5	5	3	5	5	4.42
Option 3: VPC + WAF + Shield Adv.	5	5	5	1	5	5	4.25

Prioritization

- a. SEC01-BP01 Separate workloads using accounts
- b. SEC02-BP01 Use strong sign-in mechanisms (Amazon Cognito)
- c. SEC02-BP03 Store and use secrets securely (AWS Secrets Manager)
- d. SEC05-BP01/BP02 Create and control network traffic at all layers (Amazon VPC + AWS WAF)
- e. SEC06-BP03 Implement managed services (Amazon Aurora)
- f. SEC06-BP04 Automate compute protection (AWS CloudFormation and AWS Systems Manager)

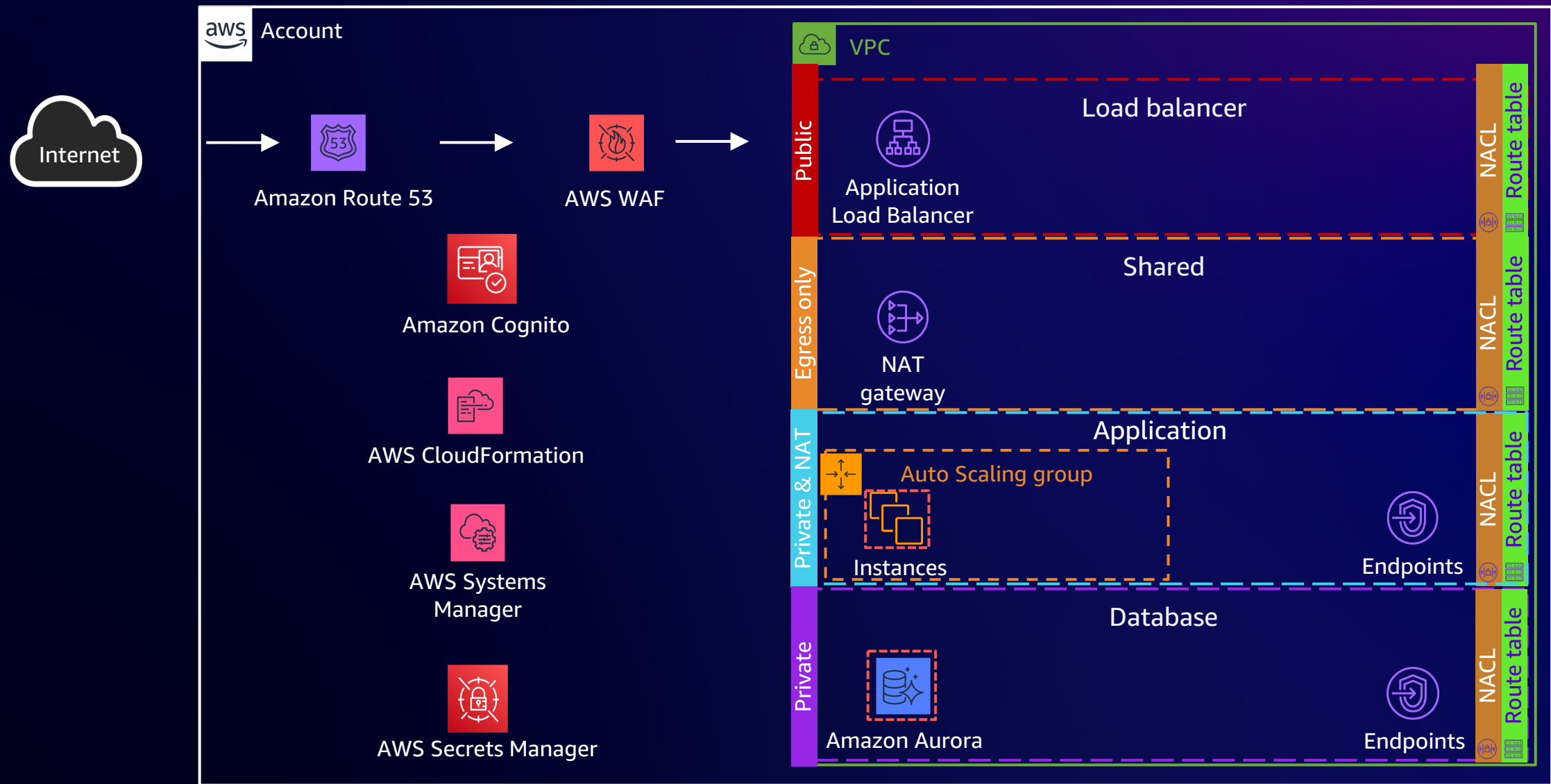


Step 3

Implement and measure



Target architecture



Takeaways

- ❖ AWS Well-Architected Framework optimization mechanism
- ❖ Mental model to navigate architectural synergies and trade-offs in cloud computing
- ❖ A semi-quantitative approach for trade-off analysis in Well-Architected Framework

Thank you!



Please complete the session
survey in the mobile app

Question list

Introduction to the Well-Architected Framework

- ❖ Can you raise your hand if you've heard of Well-Architected?
- ❖ What is the purpose of AWS Well-Architected?

AWS Well-Architected for workload optimization

- ❖ Can you raise your hand if you have conducted a Well-Architected review?
- ❖ Are you familiar with the AWS Well-Architected architectural optimization lifecycle?

Architectural trade-off analysis

- ❖ Is there anyone here who has conducted a trade-off analysis for the cloud workload?
- ❖ How did you approach it?

Backup

AWS re:Invent 2023 title and abstract

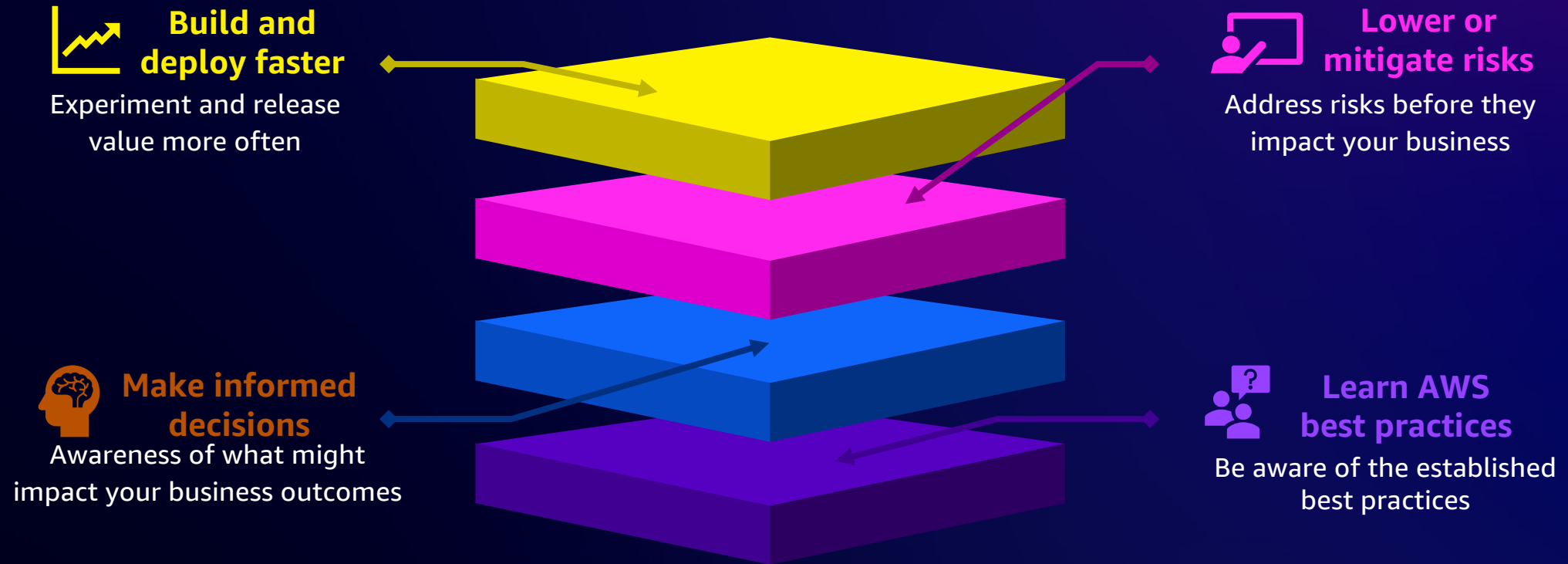
Title:

Making architectural trade-offs using AWS Well-Architected Framework

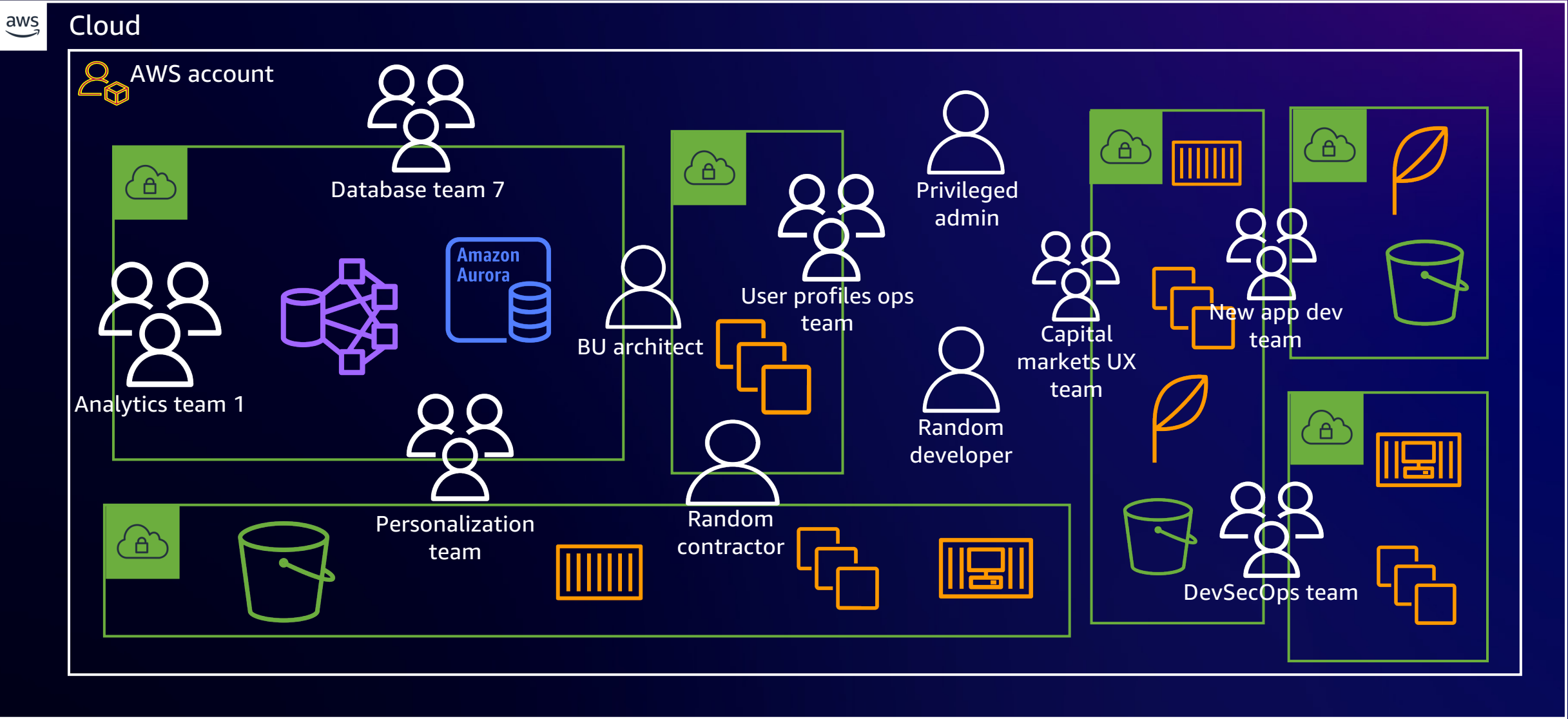
Abstract:

Technologists and practitioners who develop and deploy workloads using fast-growing cloud technologies prefer to start with established, consistent, and repeatable best practices and guidelines using the AWS Well-Architected Framework, rather than learning through trial and error. These best practices provide practical guidance on using proven architectural principles before and when designing workloads, as well as maintaining the health of a workload once it's in production. In this chalk talk, explore the impact of single and multiway trade-offs across the six pillars of the AWS Well-Architected Framework for an optimal architectural design outcome that is driven by your business requirements.

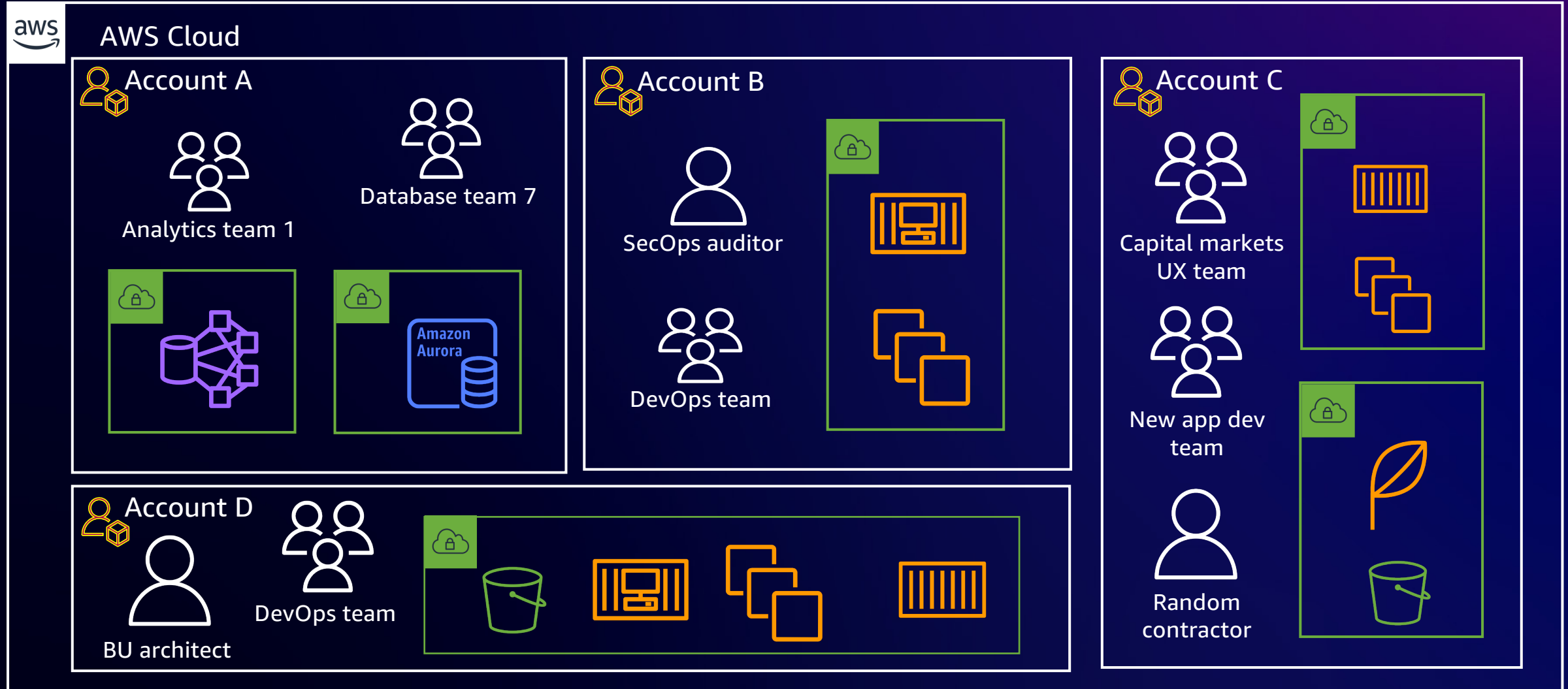
Why use the AWS Well-Architected Framework



Current architecture: AWS account overcrowding



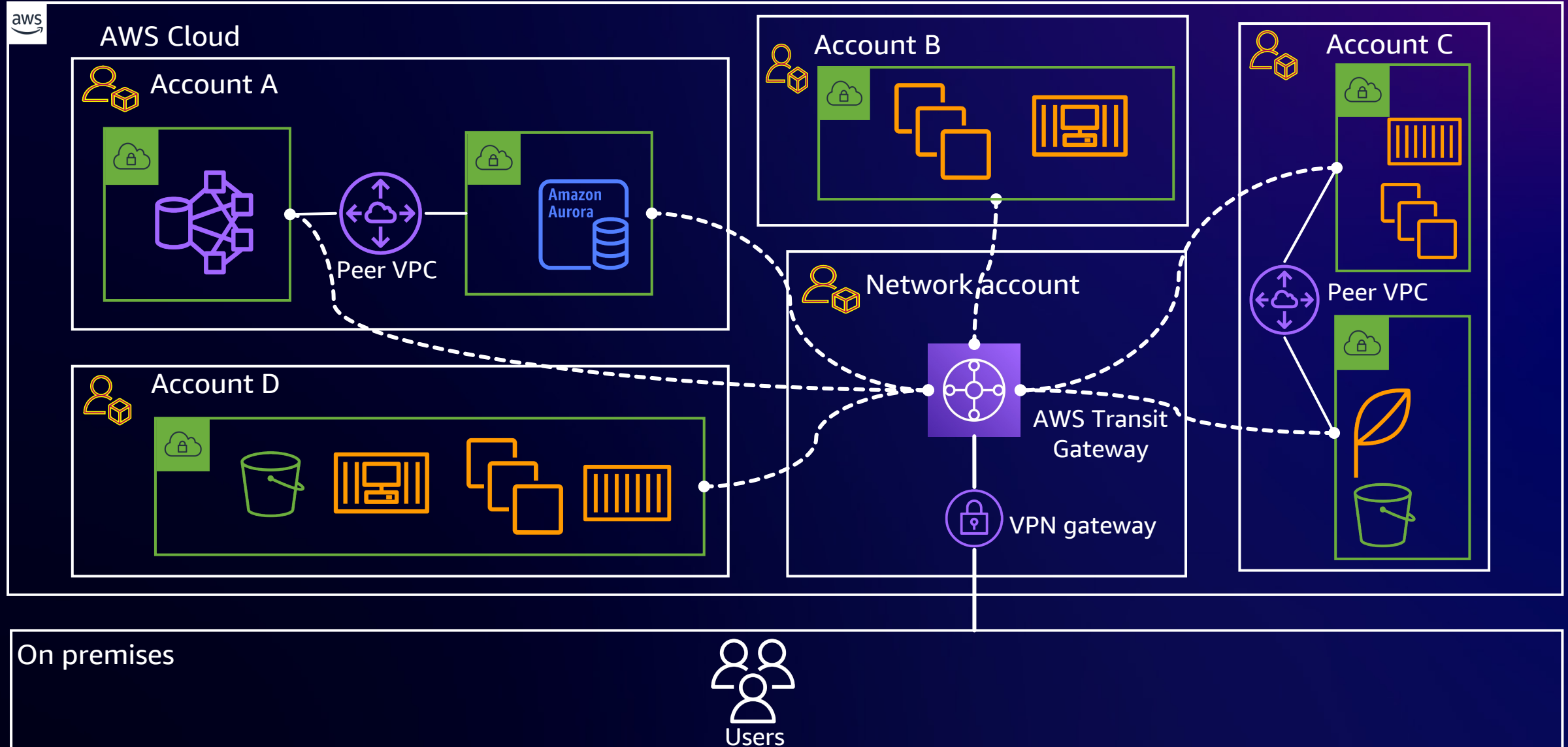
Best practices–level synergies and trade-offs



•SEC01-BP01 Separate workloads using accounts

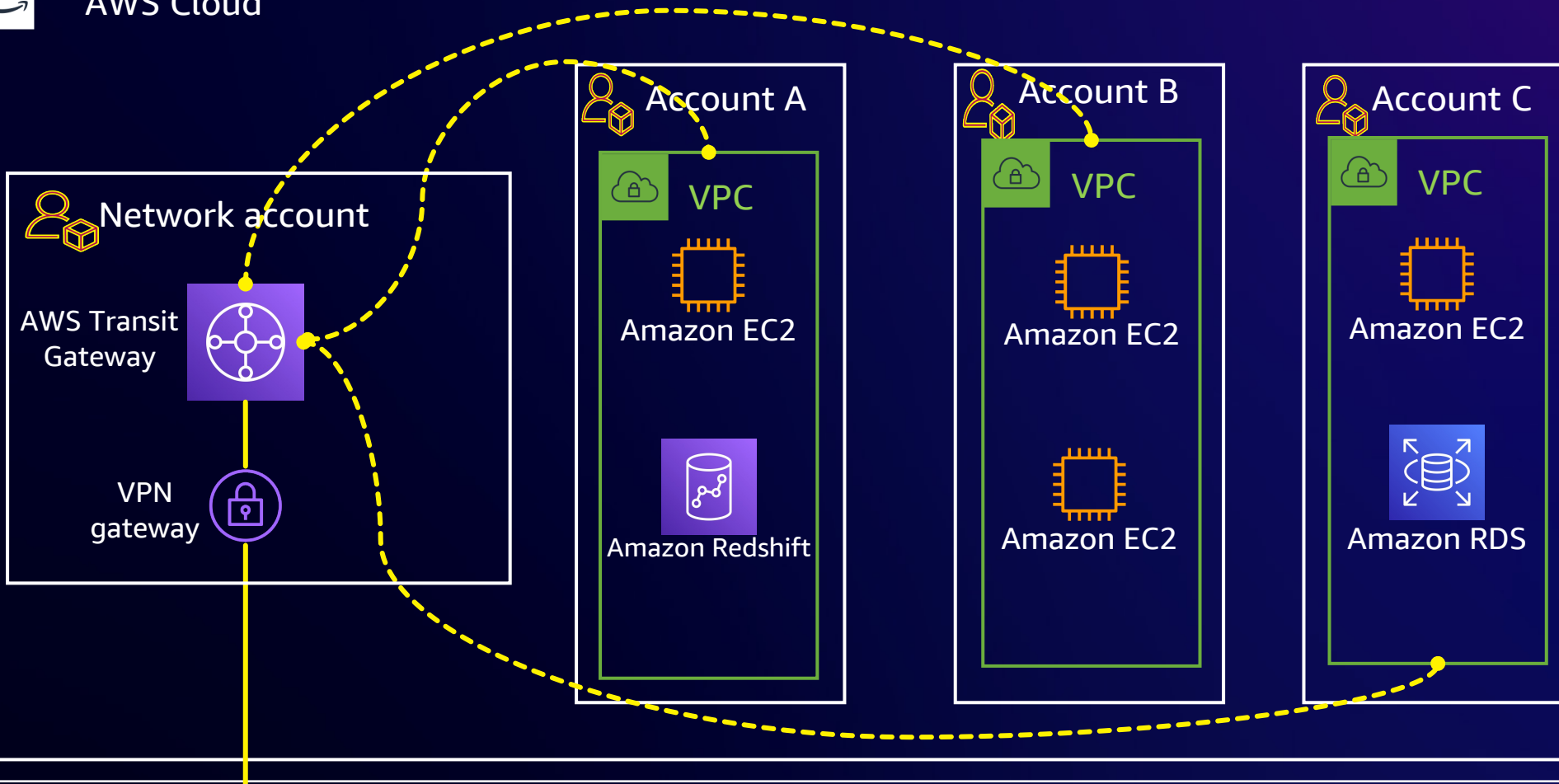
•SEC03-BP02 Grant least privilege access

Best practices–level synergies and trade-offs





AWS Cloud



On premises



Users





AWS Cloud



On premises

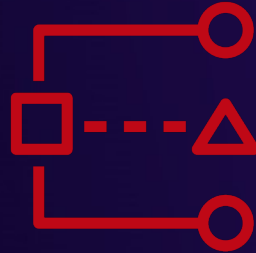


SEC02-BP01

Use strong sign-in mechanisms



Option 1
Use Amazon Cognito



Option 2
**AWS Identity and Access
Management (IAM)
and custom solution**



Option 3
Use a third-party identity provider

Assumptions

- ❖ Adding sign-in has minimal impact on latency
- ❖ Third-party solution is cheaper compared to AWS services

Trade-off analysis

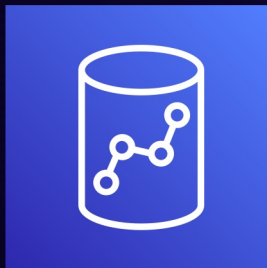
	Security (5)	Reliability (3)	Performance (3)	Cost (3)	Sustainability (1)	Ops. ex. (5)	Total score
Option 1: Amazon Cognito	5	5	3	3	3	5	4
Option 2: IAM and custom	5	3	3	1	3	1	3.2
Option 3: Third party	5	3	3	5	3	3	3.8

SEC06-BP03

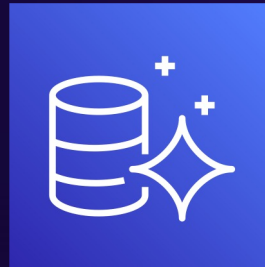
Implement managed services



Option 1
Amazon DynamoDB



Option 3
Amazon Redshift



Option 2
Amazon Aurora

Assumption

- ❖ Third-party solution is cheaper compared to AWS services

Trade-off analysis

	Security (5)	Resiliency (3)	Performance (3)	Cost (3)	Sustainability (1)	Ops. efforts (5)	Total score
Option 1: DynamoDB	5	5	3	3	5	5	4.4
Option 2: Aurora	5	5	5	5	5	5	5
Option 3: Amazon Redshift	5	5	1	1	5	5	4.6

SEC06-BP04

Automate compute protection



Option 1
AWS CloudFormation



Option 2
AWS Systems Manager